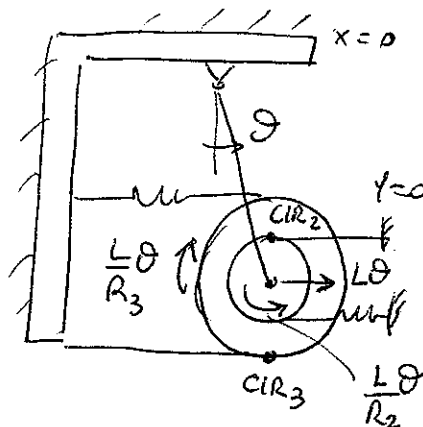


$$X = \begin{Bmatrix} x \\ y \\ z \end{Bmatrix}$$

$$\underline{Y}_{WF} = \begin{Bmatrix} x & x_A & \alpha & \beta & x_G & 0 \end{Bmatrix}^T \quad (\underline{W}_F) = \text{diag}(w_1, w_2 + w_3, J_2, J_3, w_4, J_4)$$

$$[A_{eq}] = \begin{bmatrix} 1 & 0 & 0 \\ 1 & L & 0 \\ -\frac{1}{R_2} & -\frac{L}{R_2} & \frac{1}{R_2} \\ 0 & L/R_3 & 0 \\ 1 & L/2 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$x=y=0 \quad \partial \neq 0$$



$$V_{EL} = \frac{1}{2} k_1 \Delta L_1^2 + \frac{1}{2} k_2 \Delta L_2^2 + \frac{1}{2} k_3 \Delta L_3^2$$

$$\underline{Y}_K = \begin{Bmatrix} \Delta L_1 & \Delta L_2 & \Delta L_3 \end{Bmatrix}^T \quad [K_F] = \text{diag}(k_1 \ k_2 \ k_3)$$

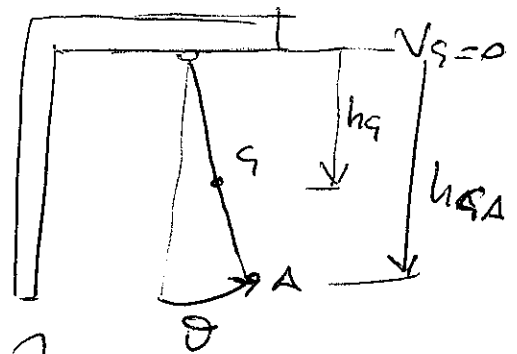
$$\begin{cases} \Delta L_1 = x \\ \Delta L_2 = 2L\theta \\ \Delta L_3 = -2x - 2L\theta + y \end{cases} \quad [A_K] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2L & 0 \\ -2 & -2L & 1 \end{bmatrix}$$

espressioni linearizzate + molle e smorzatori in parallelo $\rightarrow$

$$\rightarrow [A_F] = [A_K] \quad ; \quad [r_F] = \text{diag}(r_1 \ r_2 \ r_3)$$

$$V_G = (u_2 + u_3) g h_{GA} + u_4 g h_G$$

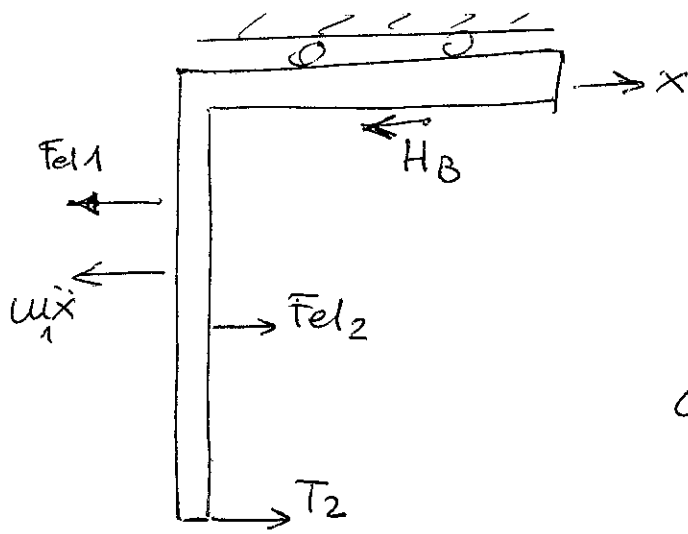
$$\begin{cases} h_{GA} = -L \cos\theta \\ h_G = -\frac{L}{2} \cos\theta \end{cases}$$



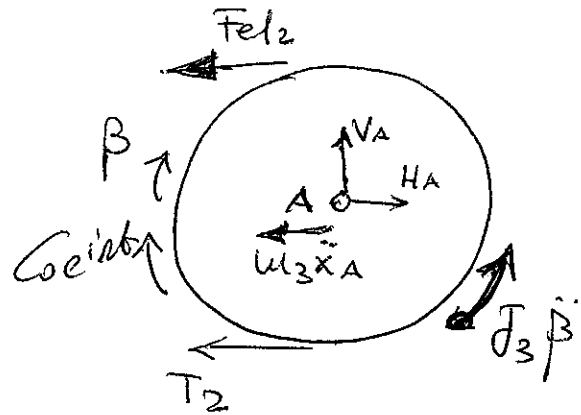
$$\rightarrow [K_G] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & (u_2 + u_3 + \frac{u_4}{2}) g L & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\delta L = C e^{i\omega t} \left(\frac{L}{R_3} \delta\theta \right) \rightarrow Q = \begin{Bmatrix} 0 \\ 4/R_3 \\ 0 \end{Bmatrix} C e^{i\omega t}$$

$\rightarrow$ partizione delle ustici $\rightarrow$ soluzione numerica



↑
sottosistema caviello



↑
sottosistema disco 3

$$\sum M_A^{\text{DISCO 3}} = 0 : \begin{cases} T_2 R_3 + C_0 e^{int} - J_3 \ddot{\beta} - \bar{F}_{el2} R_3 = 0 \\ \bar{F}_{el2} = (k_2 + i \omega r_2) \Delta L_2 \end{cases} \rightarrow$$

$$\rightarrow T_2 = (k_2 + i \omega r_2) \Delta L_2 + \frac{J_3 \ddot{\beta} - C_0 e^{int}}{R_3}$$

$$\sum F_x^{\text{cav.}} = 0 : \begin{cases} T_2 + \bar{F}_{el2} - u_1 \ddot{x}_1 - \bar{F}_{el1} - H_B = 0 \\ \bar{F}_{el1} = (k_1 + i \omega r_1) \Delta L_1 \end{cases} \rightarrow$$

$$\rightarrow H_B = T_2 + \bar{F}_{el2} - u_1 \ddot{x}_1 - \bar{F}_{el1} =$$

$$= 2 (k_2 + i \omega r_2) \Delta L_2 + \frac{J_3 \ddot{\beta} - C_0 e^{int}}{R_3} - u_1 \ddot{x}_1 - (k_1 + i \omega r_1) \Delta L_1$$